A LaTeX Showcase

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October 21, 2026

Abstract

Today we will learn about LaTeX.
This is the abstract. Here you put a short summary of your report. The tradtional structure of a scientific work is:

1. Introduction
    - The current state of the art.
    - Everything your reader should know.
    - End with the problem Statement
2. Materials and Methods (Procedure)
3. Results
4. Discussion
    - How does your findings fit the state of the art?
5. (Outlook)

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1 (Abbreviated) Basics of Scientific Writing

Main guidelines:

1. The content is the interesting part, not the formatting.
2. Plagiarism: The reader assumes everything you write is your own creation/thought. If it is not cite it.
3. Add in your mind "unless your supervisor says something different" to the end of every sentence today.

More in the individual sections: section 4, section 5
For your **thesis** read: A Manual for Writers of Research Papers, Theses, and Dissertations by Kate L. Turabian (Library code: 00/ALC 16-7) [3]

2 What is LaTeX and why use it?

Latex is a text document creation framework. It looks and feels a bit like writing computer code.
Advantages:

- Looks good out of the box
- Algorithmic layout decisions
- Formulas look nice
- Active helping community (If you cannot google it, I don't know it either)
- Some error correction
- Makes you feel like a hacker =)

Disadvantages:

- Takes time to get used to e.g.
  - \\ ends a line
  - Double spaces are ignored

Usually 3 Documents:

- .tex file (main document)
- .bib file (Bibliography)
- .pdf file

... and all the figures as separate files.
The compiler creates the pdf from the other documents.
(Show Thesis (early version))

2.1 A few Words Regarding Overleaf

- Web access.
- Most packages already installed.
- Free individual version.
- Free collaborative version is included in Sciebo.
- In the pro version you can invite collaborators.
- I like TeX Studio a bit better. (Also free, but local)

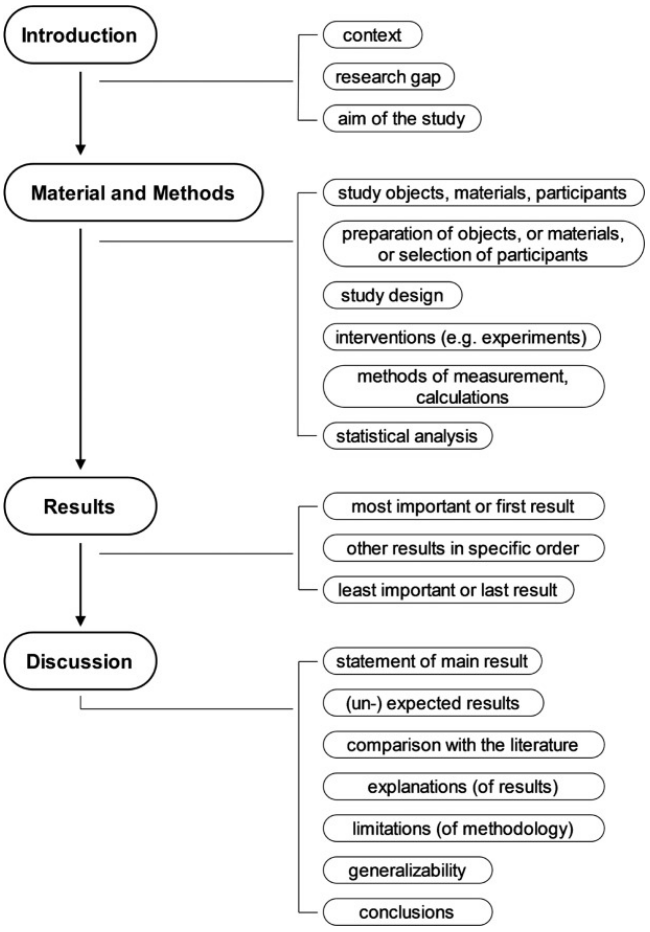


Figure 1: A guidline for students on the structure of their thesis [2]

3 Document Structure

Fabrice include the use case for Citavi here (Figure 1)!!!
Avoid having two subsequent headers without text in between.

Section

Subsection

Subsubsection

4 Figures

A figure needs a caption below the figure. The caption has to work without the text. The text has to work without the figure. A cross reference is required at a fitting position in the text (Figure 2). Avoid formulations like: "In Figure 2 we see..." or "Figure 2 shows...". Just write what is seen in the figure. Figures and tables are floating objects, LaTeX will decide where to place it. Adding a [h!] to the command will try to force to the position in the .tex document.

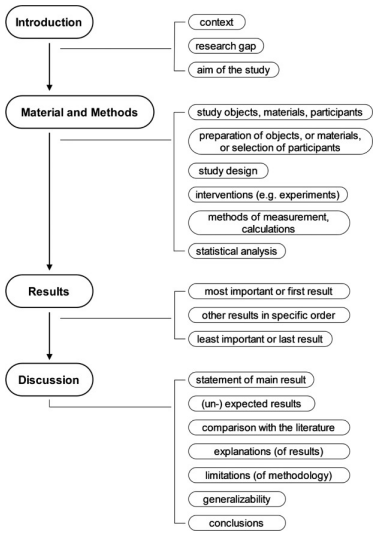


Figure 2: An example figure showing my face. Just to clarify I would have to credit my wife for the photo because I did not do it myself.

5 Tables

Tables work the same as figures. Only the caption is on top of the table (Table 1). A good table only has three lines:

1. One above the first row.
2. One below the first row.
3. One below the last row.

Table 1: Caption

header 1	header 2	header 3	header 4
1	2	6	7
3	4	9	7

6 Equations

Equations cän be σ included in different ways. In the text using single$ signs like $f(x) = (x_{2x})$. Or in a single line using double $$ signs

$$f(x) = e^{-\frac{x^2}{\sigma}}$$

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Or we use the equation environment:

$$a_{index} \tag{1}$$

$$f(x) = e^{-\frac{x^2}{\sigma}} \tag{2}$$

There are other environments for aligned equal signs

$$f(x) = x^2 + 3 \tag{3}$$
$$= 5 \tag{4}$$

Of course we can cross reference (Equation 2) or just (2)

7 Citations

We need a .bib file [4] and in the end we add commands `\cite{name}`
References only appear in the Reference list is they are used. [1] An easy way to get the bib for a source is google scholar

What is still left to be implemented is the table of **figures** and the **table** of tables.

References

[1] Fred Glover. "Genetic algorithms and scatter search: unsuspected potentials". In: *Statistics and Computing* 4.2 (1994), pp. 131–140. ISSN: 0960-3174. DOI: 10.1007/BF00175357. URL: https://link.springer.com/article/10.1007/BF00175357#citeas.

[2] Vikash Singh and Philipp Mayer. "Scientific writing: strategies and tools for students and advisors". In: *Biochemistry and molecular biology education : a bimonthly publication of the International Union of Biochemistry and Molecular Biology* 42.5 (2014), pp. 405–413. DOI: 10.1002/bmb.20815.

[3] Kate L. Turabian and Wayne C. Booth. *A manual for writers of research papers, theses, and dissertations: Chicago style for students and researchers.* Univ. of Chicago Press, 2010. ISBN: 978-0-226-82336-2.

[4] Jurriaan Wouters et al. "Lower bounds for Ramsey numbers as a statistical physics problem". In: *Journal of Statistical Mechanics: Theory and Experiment* 2022.3 (2022), p. 033211. DOI: 10.1088/1742-5468/ac5cb3. URL: http://arxiv.org/pdf/2112.11426.